Compute Platform for AI

An Open-Source Summary

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Contents

1 Lecture 1: Towards heterogeneous many-core processors

1.1 Scaling

In IC and chip design, there is two fondamental laws:

- **Denard's law:** as transistors get smaller, the power density remains constant, which leads to lower and lower supply voltage to avoid to break down the oxide due to a strong $\vec{E}$.
- **Moore's law:** every generation can fit twice as many transistors on a certain area.

1.1.1 Denard broke down

Denard's law is based on the fact that the width, length, oxide thickness and voltage is reduced by a factor α each time. This factor also influences:

- Density: α^2
- Capacitance: $1/\alpha$
- Delay: $1/\alpha$

Thus, Power = $CV_{DD}^2 f \propto 1/\alpha \cdot cst \cdot 1/\alpha = 1/\alpha^2$. Finally the power density is a constant.

On paper, this is valid but we can't infinitely scale down V or we will have lower speed the closer we come to V_{th} which is not feasible. Moreover, the wire cannot scale down as desired or we will have a bad resistance in the wire. This will make the wires a bit more "capacitive" and so the α factors will no longer cross out as Denard predicted. The power is constant and the power density is α^2 which is quite problematic.

1.1.2 Dark and Dim Silicon

The end of the Denard's scaling lead to a plateau in the power consumption of chips. We have to buy this energy efficiency. We know that the power density scales with α^2 at maximum clock frequency. So we will introduce:

- **Dim silicon:** silicon running at below max f_ϕ/α^2
- **Dark silicon:** $1/\alpha^2$ blocks that totally shut down when not used

In practice, if we scale with a factor $S = 2$ we can put 2^2 more silicon and the speed should increase by a factor 2. In *dark silicon*, we will still speedup the clock frequency but all the newly added cores will be shutdown. This is quite unsustainable as more cores (due to Adhalm's law) won't leverage from parallelism. The better idea is to not clock faster and use this extra factor 2 to produce dim silicon and then the rest with dark silicon.

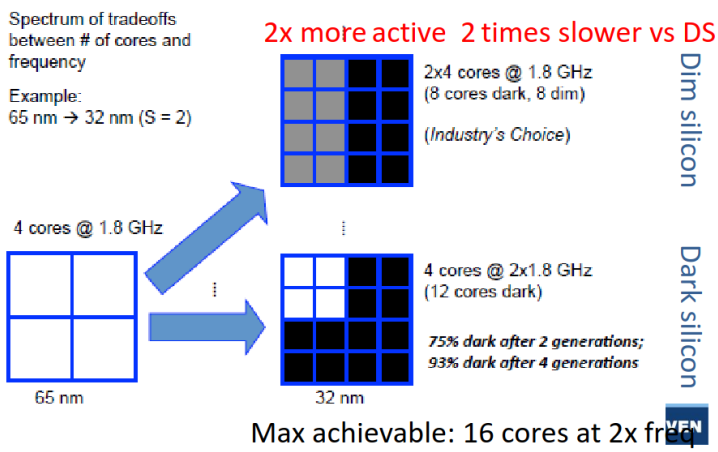


Figure 1: The two aforementioned techniques

Recently, we are using more and more dark silicon as accelerators that get turned on for specific task – accelerators.

1.2 Area for energy in single-core

To make a processor faster, we can use either:

- *Instruction-level parallelism*: VLIW, OOO super-scalar
- *Data-level parallelism*: SIMD, GPU

> Won't re-explain what those are, check the computer architecture lecture: *link*

Those are prime examples of dim silicon with lower clock speed for same throughput thanks to parallelism.

There is also the vector extension that can be seen as a sort of *temporal* data parallelism. SIMD, VLIW and other processing is often found in DSP chip since such processing is often data intensive.

1.2.1 Super-scalar

In this version, it is out of order and the processor can schedule instructions when it wants. It uses the Tomasulo's algorithm but must commit in order to avoid data dependency issues.

The advantage of such processor is the flexibility of the operations. Typically, some execution stage can take more time than others. We must use a reorder buffer (ROB) to commit in order. This is a prime example of spending extra transistors instead of clocking faster. There is a certain overhead for controlling such processors but the gains are far superior.

1.2.2 Memory area

Memory access time is the usual bottleneck in computing, a lot of time, the thread is blocked because it waits for data to arrive. Latency of cache and memory is quite important and to hide it we use multi-threading with some level of granularity to keep all cores as busy as possible.

We can't use too much of simultaneous multithreading SMT because the overhead is quite important. Need each time a program counter, register file and a reorder per thread. On top of that, the commit and scoreboard is more complex.

This is why we use various cache level and we exchange latency with transistors count. A recent trend is using large reorder buffer to see well in advance and to re-order online the instructions. This will increase parallelism and reduce off-chip memory accesses.

1.3 Area for energy through multi-core

Sadly, this is not enough and *Amdahl's law* explain that parallelism will not always lead to a speedup and lots of applications are not easily parallelizable.

1.3.1 Homogenous

We first started by doing `ctrl`+`c` and `ctrl`+`v` on the cores to realize a speedup. On top of that, we would some p-state (dim silicon) or shut it down completely (dark silicon) if not used.

1.3.2 Heterogeneous

Here, we will have dedicated low-power cores that can realize operations if low performance is needed. Depending on the workload, we can switch to efficiency core or performance core. This is present in the **big.LITTLE** ARM architecture. We must use the same instruction set to seamlessly transfer from one core to the other.

This idea is heavily used in embedded devices like smartphone and is starting to make its way through in personal computer, typically Apple's computer.

1.4 Area for energy through domain specific accelerators (DSA)

We are now facing the utilization wall. More and more silicon must be dim or dark, we must use the extra transistors for something. Here comes accelerators which are custom ASIC blocks for specific function.

2 Lecture 2: ISP's, GPU's and AI Workloads

2.1 ISP's & GPU's for graphics/images

2.1.1 ISP applications, architecture and operation

To take a picture, many steps are required: calibration, RGB conversion, encoding, post-processing, … Image Signal Processor is one of the earliest example of ASIC. The operations are fixed and easily paralleziable.

Initially, it was mostly hardwired but more recently introduction of more flexibility. It gained popularity in commercially available ISP as people demanded more from their smartphones and cameras.

2.1.1.1 IPU This is becoming more and more like an Image Processing Unit where we want the best of both world:

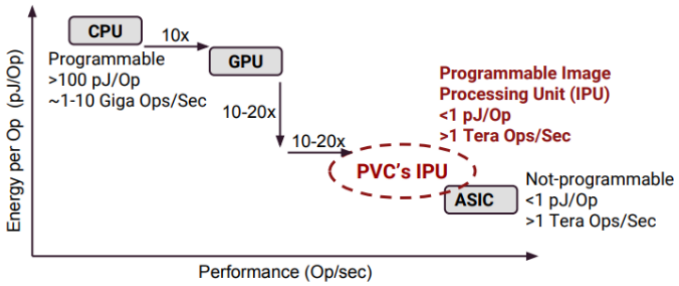


Figure 2: IPU

This is the idea introduced in the google pixel family with their Pixel Visual Core that allows data processing alongside the CPU. They use multiple small ASIP and use their locality on the die to exchange information between each other. It is a fast paced processing where data is consumed and sent to the next processor.

They arange multiple Processing Elements (PE) next to each other and hardwired them all together. They are composed of ALU's and MAC units and can be programmed and sent to the next PE. The dawn of NPU…

In short, the goal is to exploit **parallelism** and functional pipelining. We try to provide more programmability without losing in efficiency.

2.1.2 GPU applications, architecture and operation

> Repeat of Computer Architecture, Summary here.

2.1.2.1 Mobile Here we will try to maximize throughput under power and area limitation. We will also use extensively lower precision data types that can relieve stress and increase throughput while minimizing its impact on the output quality.

We will more and more see tight integration with the CPU and GPU with shared coherent memory space. More and more programmability with CUDA or openCL. Anything that can be parallelized is interesting to run on a GPU.

2.2 Deep learning algorithms

> **Definition of AI**
>
> Any program that mimics human intelligence. A program that can sense, reason, act and adapt

2.2.1 What is deep learning?

- **Machine learning**: is a subset of AI and it is an *algorithm that is able to learn from data*.
  - **Deep learning**: a representational machine-learning using multi-layered neural networks (NN)

The idea of a deep-learning network is one that has many layers and the computer truly learns on its own without much human interaction or tuning. It learns its own representation of the world or task he must accomplish.

2.2.2 From neural nets to deep neural nets

A basic neural network is composed of many **nodes** which forms a **layer**. They are often *fully-connected* to the next layer. The NN will adjust its weight based on algorithm during the training phase. Weight represents how much the information of one node will be forwarded to the next one. The AI can also tweak the **bias** to adjust its performance. Finally, to reduce the "*fuzziness*" we use some **activation functions**.

$$o_n^l = \sigma\left(\sum_m w_{m,n}^l \cdot o_m^{l-1} + b_n^l\right)$$

Table 1: Various σ activation function

Name	Function	Good in math	Easy in HW
Sigmoid	$\frac{1}{1+e^{-x}}$	**V**	**X**
Tanh	$tanh(x)$	**V**	**X**
ReLU	$max(0, x)$	**X**	**V**
Leaky ReLU	$max(0.1x, x)$	**X**	**V**
Maxout	$max(w_1^T x + b_1, w_2^T x + b_2)$	**X**	**V**
ELU	$x \geqslant 0; x$ else $\alpha(e^x - 1)$	**X**	**V**

By using labeled data, we can train the AI to accomplish a task. Using the AI to do for example pattern recognition is called *inference*.

2.2.3 Classes of deep neural networks:

2.2.3.1 DNN In the simple example of NN, we had a vector as an input. But, we can also have an image (matrix) as the input which we can apply the same pattern. This time, we will use a matrix notation to realize 2D-2D operations; a fully connected layer looks like:

$$o_{nx,ny}^l = \sigma\left(\sum_{mx,my}^{M,M} w_{mx,my,nx,ny}^l \cdot o_{mx,my}^{l-1} + b_n^l\right)$$

The matrices are quite large and this can be detrimental sometimes as one result is based on the full image.

2.2.3.2 CNN We are no longer fully connected, it is a sparse matrix. It has better convergence and faster training.

$$o_{nx,ny}^l = \sigma\left(\sum_{fx,fy}^{FX,FY} w_{kx,ky}^l \cdot o_{xn+kx,ny+ky}^{l-1} + b_n^l\right)$$

This will avoid to have a M matrix but rather a small kernel that is sliding over the matrix and of size $Fx \cdot Fy$. We can go further and perform some tensor kernel to apply the same or different kernel on multiple inputs. Each inputs can influence the same output. A kernel is of size $Fx \cdot Fy \cdot C$.

$$o^l_{nx,ny,k} = \sigma\left(\sum_{fx,fy,c}^{FX,FY,C} w^l_{kx,ky,c} \cdot o^{l-1}_{xn+kx,ny+ky,c} + b^l_n\right)$$

This operation requires 7 nested for loops. Bad news for software engineer, good news for hardware engineer has it opens the door for massive parallelism.

To avoid the size to get too big, we often finish by a **Pool and normalization** layer to "compress" the info. Finally we inject it in a classic fully connected NN to output a scalar result.

There are many architectures and flavor of CNN, each with their pros and cons, trying to solve different problems.

DO THE DEPTHWISE POINTWISE CONV EXPLANATION

2.2.3.3 Transformer and LLM The secret sauce of current LLM's is the use of the **attention** mechanism. It is an algorithmic representation of linguistic property of words and sentences. It will connect words between each other regarding the context which allows to not only process one word but also its context surrounding it. The maximum path length is only $\mathcal{O}(n)$ as the information is already embedded in the word after the attention process.

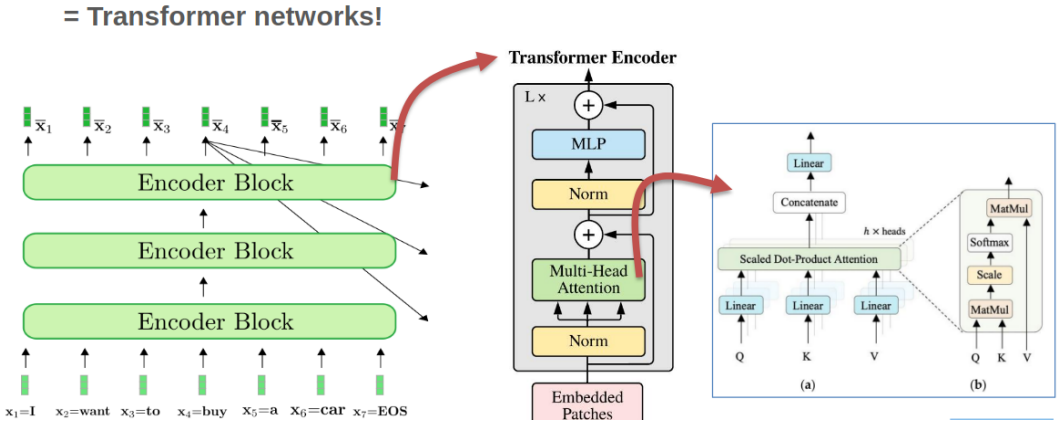


Figure 3: Transformers encoder

The query is for one word we want to inspect, the keys are the result of the query. This form a linear transformation which will tweak the high-dimensional embedding space to "distort" and bring words that are alike closer together. Finally we use the value matrix that will realize a linear transformation to help and find the next most probable word.

$$softmax(\frac{\underline{x}Q_w * XK_w^T}{\sqrt{100}}) * XV_w$$

Multiplication can be quite annoying but with efficient hardware, it can be accelerated. The real bummer here is the $softmax$ function:

$$softmax = \frac{e^{x_i}}{\sum_{j=1}^{K} e^{x_i}}$$

Where x_i are values. The issue is that it can only be computed after processing all of the data. It is also a non-linear function that cannot be realized in one single pass. This means that we must re-access value from the cache which is slow and **not** energy efficient. Same story for the $norm = \frac{value_{original} - \mu}{\sigma}$.

After doing this pre-fill and encoding, we want to generate some new tokens. This is handled by the decoder block.

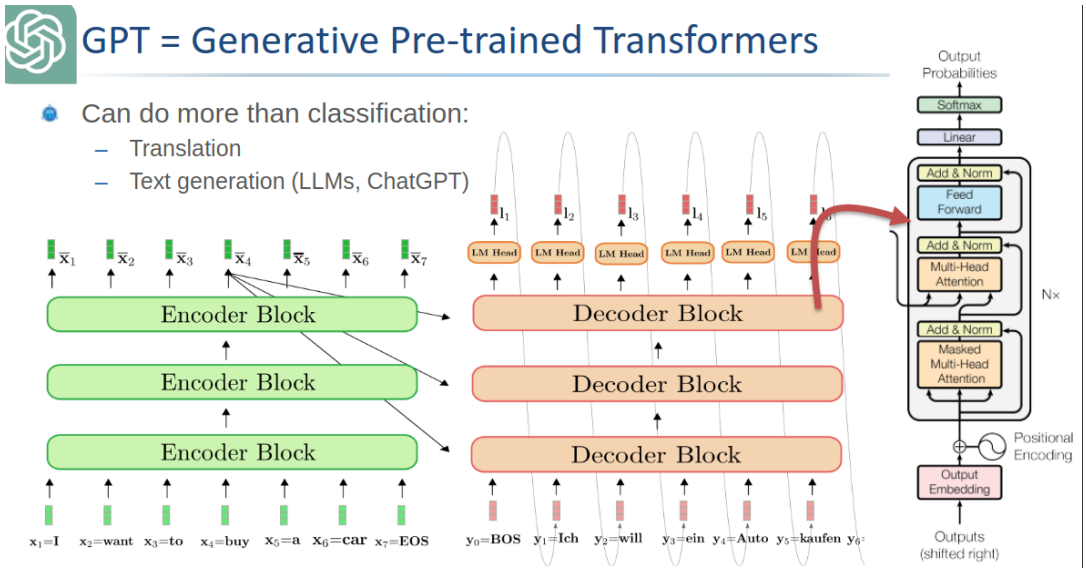


Figure 4: Full architecture

Those matrix multiplications are quite repetitive, we can thus save the result in what we call **KV cache**. We only append the new token and perform some vector matrix computation instead of the full matrix matrix computation.

- Prefill: compute limited
- Decode: memory limited

On a side note, most LLM's are now *decoder-only* but we still have this prefill stage.

3 Lecture 3: Efficient Deep Inference: Hardware Bottlenecks & Parallelization

3.1 Baseline and hardware challenges

The two main efficiency metrics are:

- Latency: time per inference, time to get the first token
- Throughput: inferences per second

We often represent operations per second as TOPs - 10^{12} operations per second. When batching (running multiple queries at once) we gain in efficiency and it provides better throughput as we can reuse the weight of layers loaded in RAM for other users at the same time.

We also look at the energy and power. Where, TOPs/W (or TOP/J) matters depending of the scenario (cooling, embedded, infrastructure, … constrained)

With Moore's law, we should get twice the efficiency roughly every 2 years. The issue is that AI's complexity is getting more and more complex at a higher rate leading to a massive gap between computation abilities and needs.

We can fight at multiple font:

- TP/Latency: $time/op \cdot 1/utilization \cdot ops/inf$
- Energy: $energy/op \cdot 1/utilization \cdot ops/inf$

3.1.1 Metrics for execution efficiency

Matrix multiplications are needed in prefill stage. We call this operation a General Matrix Multiplication or GeMM. This type of optimized operations are implemented in the BLAS library that ensure fast and smart computation. A CPU has small **Processing Element (PE)** with specific datapath to accelerate such calculations.

The main drawback is the recurrent movements of data from on-chip to off-chip. A typical NPU will try to optimize those steps. Sadly, we will hit the memory wall which will lead to unavoidable latency and energy constraints.

One solution is to use lower precision operations to reduce the energy cost. Thankfully, energy savings is linear with the precision but the quality of the LLM isn't, especially if we use smart algorithm.

3.1.2 A tale of two rooflines

The roofline diagram we are familiar with is the $AI - TOPs$ one. Where we plot the Arithmetic Intensity as Operations per bytes of memory access. We have one horizontal line where there is the maximum compute in TOPs N_{op} and a diagonal line that is $AI \cdot BW$ depicting the bottleneck of the memory access. The ideal operating point is where those two line crosses as we are using the maximum of memory and compute. Otherwise the maximum attainable performance is $min(AI \cdot BW, N_{op})$.

But we can also have the **energy roofline**. We wil have the horizontal line being the minimum for an operation (without the transfer and other operations) and the the energy per DRAM access that is AI/E_{DRAM} access. We have the attainable efficiency as

$$\text{Attainable efficiency} = \frac{1}{E_{comp} + E_{mem}/AI}$$

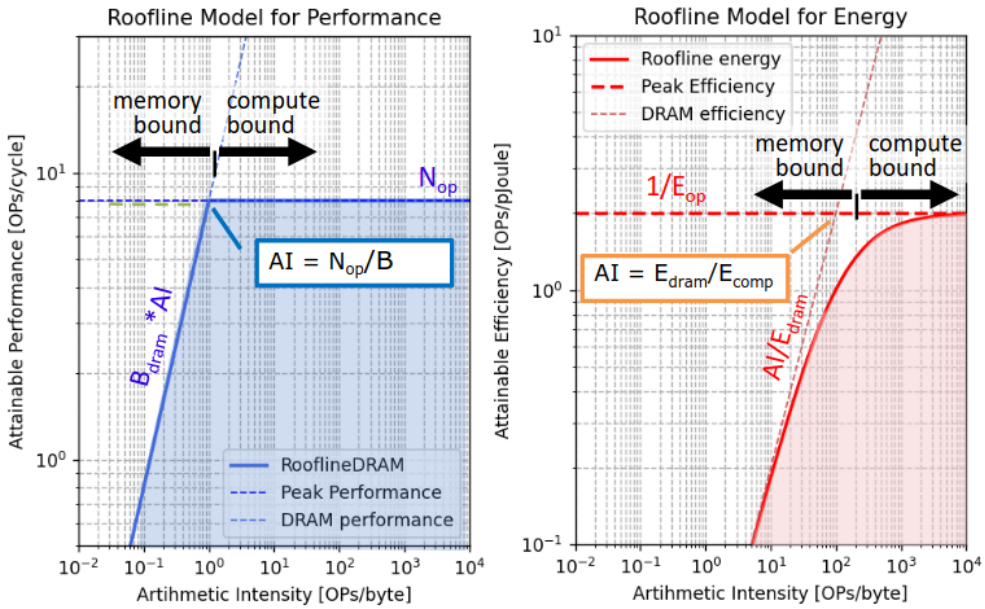


Figure 5: The rooflines diagram

We can see that the optimal performance point may not be the most energy efficient one! It all depends on our goals and seeing the large scale AI infrastructure we prefer to be slightly compute bound to avoid excess energy overhead.

3.2 xPU processor enhancements for AI

As explained earlier, the extra transistors we are gaining thanks to Moore's law need to be used to something. That's why we choose to do specific ASICS block or xPU. They all rely on one of the 5 tricks explained in the coming subsections.

Every trick here will try to push some limits (AI, BW, ...) to further improve the performances.

3.2.1 Parallelization (spatial unrolling optimization)

3.2.1.1 Parallelization in CPU and GPU cores Sadly, simply parallelizing work won't reduce the AI or increase it. We need to also push the peak performance or else no gain.

This is why we are interested in SIMD (parallel MAC in FP), Super-scalar (parallel load/store and compute) and also multi-core parallelism.

A good solution that is often employed is **fused multiply add** to avoid the extra load/store. Multiple inputs for one output, we almost divide memory access by 2.

Basic GPU's do not have this fused MAC in the classic CUDA cores, only massive parallelized MAC.

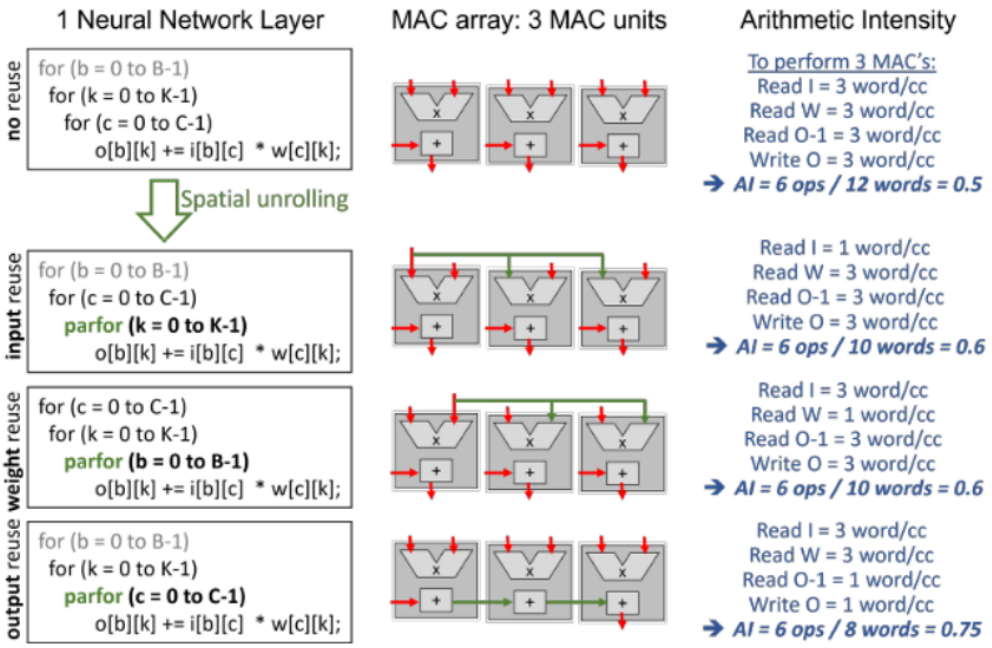


Figure 6: The goal is to reuse data to reduce AI

3.2.1.2 Advanced spatial data reuse in NPU's & Tensor Cores Of course, if we could unroll and reuse everything it would be ideal but the values are getting quickly out of hand. For example, with $B = 9; C = 4; K = 8$ where $B \times C$ and $C \times K$ are the inputs and weights dimensions respectfully. We thus need $B \cdot C \cdot K = 288$ MAC to do all the computations all at once. For the amount of access we have $K \cdot C + B \cdot C + B \cdot K = 8 \cdot 4 + 9 \cdot 4 + 9 \cdot 8 =$.

But we can't have such high number of mac even for those simple architectures. We often have only $100 - 1000$'s MAC in an accelerator. We will smartly fold convolutions on multiplier array. The main optimization criterion will be to optimize towards minimal memory access.

A good way is to mix spatial and temporal unrolling. For example do sub-vector sub-vector multiplication. The compiler can also optimize the loops and re-organize (tiling, ...), this is **dataflow optimization**.

```
for (b = 0 to B-1); // for each image in the batch
  for (k = 0 to K-1); // for each output channel
    for (c = 0 to C-1); // for each input channel
      o[b][k] += i[b][c] * w[c][k];

// Compiler optimization - tiling + spatial optimization
for (b2 = 0 to B/S-1); // for each image in the batch
  for (k = 0 to K-1); // for each output channel
    for (c = 0 to C-1); // for each input channel
      parfor (b1 = 0 to S-1); // for each image in the batch
        o[b2*S+b1][k] += i[b2*S+b1][c] * w[c][k];
```

In this code example, we are dong *weight reuse* as we the weight are reused (the indices are not impacted by a parfor loop so only 1 will be loaded at a time). If we insert the parfor in any other lines we will do some input or output reuse.

Table 2: Comparison of techniques, S = spatial reuse factor

	Weight reuse	Input reuse	Output reuse $\rightarrow$ FMA
Weight BW	1	S	S
Input BW	S	1	S
Output BW	$S+S$	$S+S$	$1+1$
AI	$2S/(3S+1)$	$2S/(3S+1)$	$2S/(2S+2)$

3.2.1.3 State of the Art examples

The real magic sauce of NVIDIA is their tensor cores. The major speed up comes from complex instructions such as HMMA and IMMA (Matrix Multiply Accumulate). It allows us to work with tensor and realize such operation efficiently. According to NVIDIA, such operation only has a $16-22\%$ of overheard compared with scalar or vector mac which can have up to 2000% of overhead.

Most of tensor cores in a NVIDIA chip are quite "small" they realize operation on 4x4 tiles to increase utilization.

For a 4x4 we have 128 operations and 16 input, weight access and 16 output read and access totaling at 64 memory access. The arithmetic intensity is $128/64 = 2$. Of course, we have to do tiling for bigger matrices, the code can look like:

```
for (b2 = 0 to B/PB-1);
  for (k2 = 0 to K/PK-1);
    for (c2 = 0 to C/PC-1);
      parfor (c1 = 0 to PC-1);
      parfor (k1 = 0 to PK-1);
      parfor (b1 = 0 to PB-1);
        o[b2*PB+b1][k2*PK+k1] += i[b2*P+b1][c]*w[c][k2*PK+k1];
```

Tensor cores give a 10x boost to operations. Other improvements are such as larger kernel, support for other data type. Flexibility and shared architecture for FP16 and INT4.

3.2.1.4 Tesla NPU

Instead of a 3D approach, they do a classic GeMM operation:

```
for (c = 0 to C-1);
```

```
  parfor (k = 0 to 95);
  parfor (b = 0 to 95);
    o[b][k] += i[b][c] * w[c] [k];
```

We must have a large bandwidth to process all of this together. This is quite large and only a vertical company can choose this solution. Since NVIDIA must support multiple form of workload, they cannot bet that everyone will use 96x96 matrix multiplication but Tesla can make this and reduce underutilization.

3.2.1.5 Huawei DaVinci

```
for (b1 = 0 to B/16-1); //for each image/pixel in the batch
  for (k1 = 0 to K/16-1); //for each output channel
    for (c1 = 0 to C/16-1); //for each input channel
      parfor (b2 = 0 to 15); //for each image in the batch
      parfor (k2 = 0 to 15); //for each output channel
      parfor (c2 = 0 to 15); //for each input channel
        o[b][k]+= i[b][c] * w[k][c];
```

But which is better for 1024 MAC's ? 2D or 3D ?

- 2D: we will have to re-access 32x32 output making the total access cost $32 + 32 + 2 * 32 * 32$ For $2 * 1024$ which results in a total AI of $2048/(2 * 32 + 2 * 1024) \approx 1$.
- 3D: we will have 8x8x16 MACs with $2(8 * 16) + 2 * (8 * 8)$ memory access for the same amount of operations leading to $2048/(2 * 128 + 2 * 64) \approx 5.4$.

In this case 3D is much better, but is it always ?

3.2.1.6 Multi-core parallelism Use the `parfor` loops at a higher level not just the datapath! We must communicate between core and split data among core until gathering all the results. It is **sharding**.

```
parfor (b = 0 to B-1); // Unrolling at the multi-core level
  for (k = 0 to K-1);
    parfor (c = 0 to C-1);
    o[b][k] += i[b][c] * w[c][k]; // Spatial unrolling at the core level
```

Table 3: Distributing and parallelizing

Options	Input	Weight	Output	Comment
Data parallelism	User data split on different GPU's	Replicated	No gathering needed	Everything is working in parallel no need to gather at the end
Tensor parallelism	Replicate the input across	Distributed (row-wise)	Gathering	We slice the tensor and each GPU's realize part of the MMA and then recombine all
Tensor parallelism	Replicate the input across	Distributed (column-wise)	Gathering	Like tensor parallelism but in the other dimension

Options	Input	Weight	Output	Comment
Pipeline parallelism	All input go to the same GPU's	Set of weight per GPU's	Output are grouped together (unicast)	We distribute the weights across GPU's, can lead to underutilization as not every layer needs this much operation compared to other

There is not just one perfect technique, most of the time, we must combine different techniques depending on our need, architecture, FoM, ...

SambaNova startup is also believing in parallelism with intelligent on chip data routing for efficient distribution of computations.

At the end, we all try to raise the roof and tend to higher arithmetic intensity to gain in computing power.

4 Lecture 4: Efficient Deep Inference: Sationarity & Tiling

4.1 Stationarity (temporal unrolling optimization)

As always our goal is to improve the Arithmetic intensity to:

1. Improve performances
2. Improve energy efficiency

4.1.1 Tiling in CPU and (traditional) GPU cores

If we don't have data locality (e.g.: systolic array), we can still improve the operation by fetching one line of data and reusing across. Typically we fetch one row of A matrix and all the columns of the B matrix to obtain the first row of results. This is called **tiling**.

We can also be aware of the abstraction level of the memory and to reuse what was fetched in SRAM or other caches. For example, if the computer fetches a full row of 4 by X of matrix A, then we could compute all of those rows first.

Of course, this makes the AI roofline plot tougher has we have distinct bandwidth for each memory type.

$$\min(BW_{dram} \cdot AI_{dram}, BW_{sram} \cdot AI_{sram}, BW_{rf} \cdot AI_{rf})$$

With the temporal utilization:

$$\text{Temporal Utilization (TU)} = \frac{\text{Compute CC}}{\text{Total CC}}$$

For the energy efficiency, we take the weighted sum of energy access of DRAM, SRAM and register file. Not really clear to what happens at lower level.

The preferred stationarity is the output sationarity as we need 1 output read and 1 output write. Outputs are often 4 times larger than inputs !

```
for (b = 0 to B-1); // for each image in the batch
  for (k = 0 to K-1); // for each output channel
    for (c2 = 0 to C/P-1); // for each input channel
      parfor (c1 = 0 to P-1);
        o[b][k] += i[b][c2*P+c1] * w[c2*P+c1][k];
```

With T the factor of data reuse we can establish for matrix and vector computation the following results:

Table 4: Matrix computation efficiency for various temporal reuse

	Weight-stationary	Input-stationary	Output-stationary
Weight BW	S/T	S	S
Input BW	S	S/T	S
Output BW	1	1	1/T
AI_SRAM	2S/(S/T+S+2)	2S/(S+S/T+2)	2S/(2S+2/T)

Table 5: Vector computation efficiency for various temporal reuse

	Weight-stationary	Input-stationary	Output-stationary
Weight BW	1/T	1	1
Input BW	S	S/T	S
Output BW	S	S	S/T
AI_SRAM	2S/(1/T+3S)	S/(1+S/T+2S)	S/(1+S+2S/T)

4.2 Advanced temporal data reuse (stationarity) in NPU's & TC's

- Huawei DaVinci core
  - Based on tensor architecture with output reuse. The AI is 8 using 4096 MAC.

```
for (b1 = 0 to B/16-1); for each image/pixel in the batch
  for (k1 = 0 to K/16-1); for each output channel
    for (c1 = 0 to C/16-1); for each input channel
      parfor (b2 = 0 to 15); for each image in the batch
      parfor (k2 = 0 to 15); for each output channel
      parfor (c2 = 0 to 15); for each input channel
      o[b][k]+= i[b][c] * w[k][c]
```

- Tesla NPU
  - It has roughly 10.000 MACs and only fetch 2*96 weights at each cycle bringing the AI to 104.

```
for (c = 0 to C-1);
  parfor (k = 0 to 95);
  parfor (b = 0 to 95);
  o[b][k] += i[b][c] * w[c] [k];
```

- Systolic arrays: Google TPU & ARM's systolic tensor array
  - Pass inputs and outputs to neighboring PE. We pre-load all the weights in the array and we pass partial sum to the next PE to complete the operation. Lots of register switching!
  - 256 by 256 for variable size B*256 input, it requires only B clock cycles with an AI of 256.

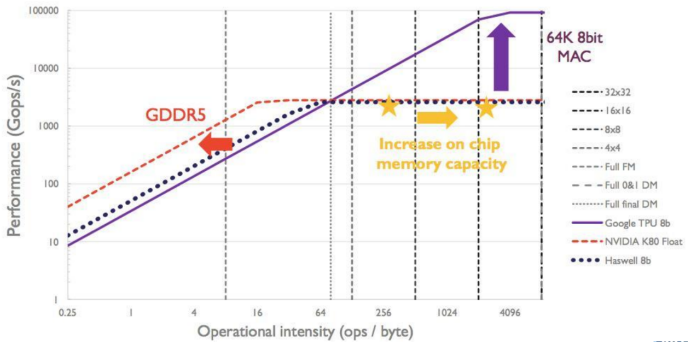
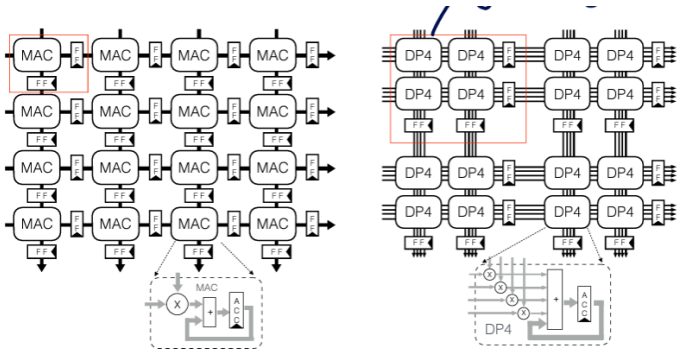


Figure 7: Comparing the solutions

The biggest issue with systolic array is the register overhead.



(a) Systolic Array (SA) 1×1×1_4×4 (b) Systolic Tensor Array (STA) 2×4×2_2×2

Figure 8: Solution to reduce overhead by 20%

5 Lecture 5: Efficient Deep Inference: Quantization

Using quantization will result into better roof as we can push the throughput further with low-accuracy operation using the same hardware. Instead of using full precision FP32, we can use lower bit precision as 4-bit, 1-bit, … Lower order precision is often computed in `int`

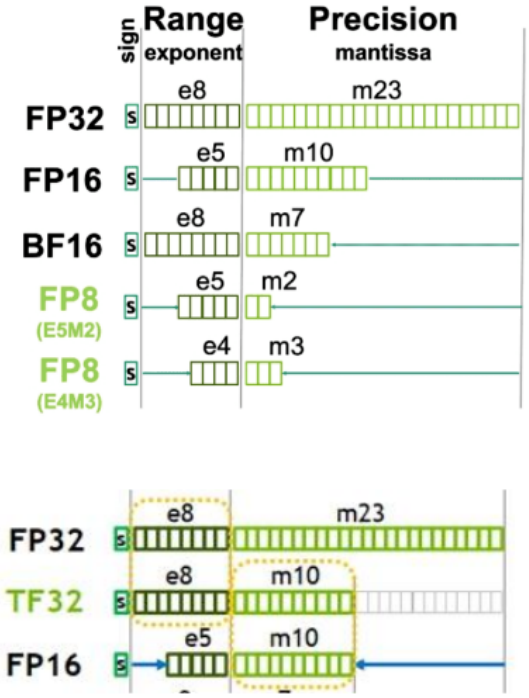


Figure 9: Datatypes for quantization

The TF32 format is used inside Nvidia's card in memory. It is not a real 32 bit float. They use this format as doing fp32 fp32 operation will unavoidably lead to trimming and rounding errors.

5.1 Block quantization

We can group exponent together and then use the mantissa for the operation. It is a form of **dynamic** fixed point arithmetic. The exponent is fixed and the mantissa can still vary. This allow for smart operation that can re-use previously computed results.

We can push this idea further by trying to group more than just the same components together. We can group multiple numbers under the same scale. This is the basic idea behind int quantization:

$$w_{quant} = quant(S_q \cdot (w - O_q))$$

We must add an offset O_q in the case the distribution is not symmetric.

Table 6: The MX-compliant data types

Format Name	Block Size	Scale Data	Scale bits	Element data format	Element bit-width
MXFP8	32	E8MO	8	FP8 (E4M3 / E5M2)	8
MXFP	32	E8MO	8	FP (E2M3 / E3M2)	6
MXFP4	32	E8MO	8	FP4 (E2M1)	4

Format Name	Block Size	Scale Data	Scale bits	Element data format	Element bit-width
MXINT	32	E8MO	8	INT8	8

5.1.1 PQT and QAT

The cheapest way to implement quantization is to do some **Post-training quantization**. After training, an expensive and time consuming step, we apply the quantization on the weights. We need some calibration data to make sure the quantization will not impact negatively the model.

On the other hand, **Quantization aware training** is ran at training. We quantize on the fly and and finetune with the training data. This is often better but more costly especially with LLM's.

Figure 10: PQT vs QAT

5.1.2 Mixed precision NN

In the same idea as quantization per group, we can apply different quantization order in the neural network. During the training, all possibilities exist and then model will adapt the π transfer function. Finally certain layers will prefer different quantization order than other.

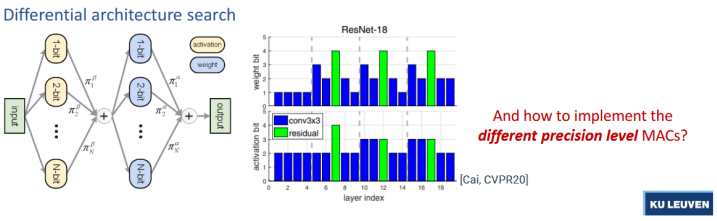


Figure 11: Differential architecture search

5.2 Variable precision `int` MAC's

All of this is cool and interesting but if we can't reuse hardware or do something a bit smarter, we cannot leverage from those algorithmic technique in real life. This is where versatile MAC arrays come into play.

5.2.1 Data gating

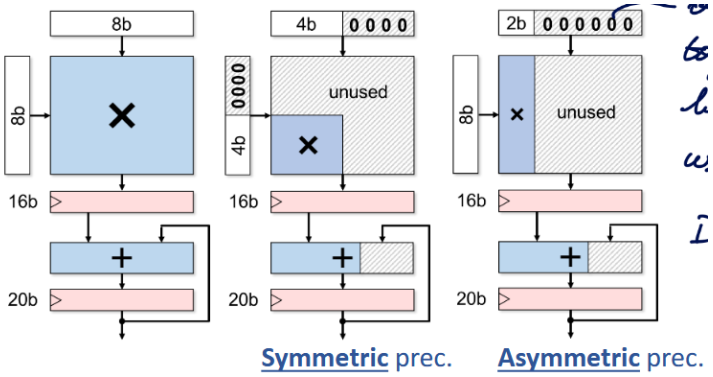


Figure 12: Data gating MAC

It is a relatively naive idea and will let lots of space unused ! We gate the LSB's to avoid unnecessary toggling and reduce energy consumption.

5.2.2 Multi-gating

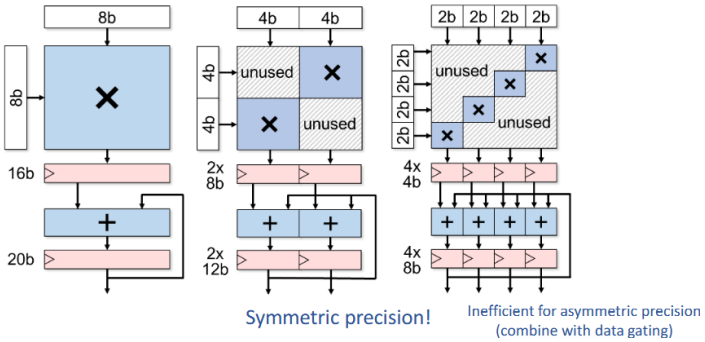


Figure 13: Multi-gating MAC

This can only be done for **symmetric** operations as asymmetric is quite inefficient. Interestingly, the more sub-word we have, the larger are the registers getting. It is not depicted in the figures, but we must also ensure some gating of the signal to avoid carry propagation between each sub word.

5.2.3 Add/shift-gating

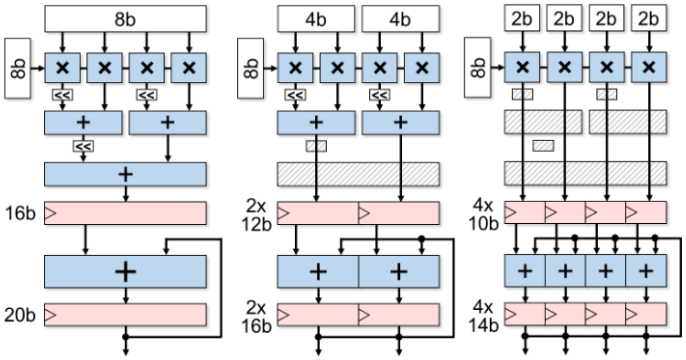


Figure 14: Add/shift-gating MAC

This method is suitable for asymmetric operations. We operate on sub-unit and combine the results as required. This method presents a finite granularity and the more range we want to support the more adder need to be chained hurting the critical path.

5.2.4 Bit-serial way

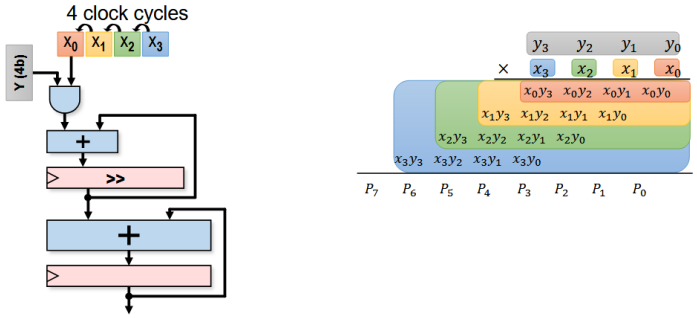


Figure 15: Bit-serial way MAC

We chain operations depending on the width and adjust the clock cycle.

5.2.5 Comparing

At full precision, data-gating is the best as there is little to no extra hardware or register toggling. But if we look at a reduced precision bit-serial and add-shift have the best energy usage and the best hardware usage.

5.2.6 From MAC to array level

For fair comparaison, we should compare at the MAC array as it presents many data sharing opportunities.

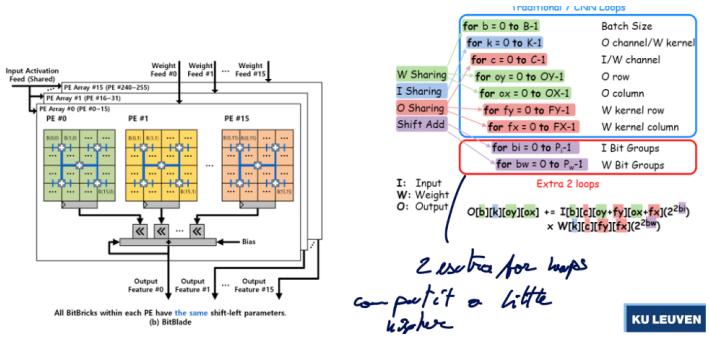


Figure 16: Data sharing can be seen as 2 extra for loops

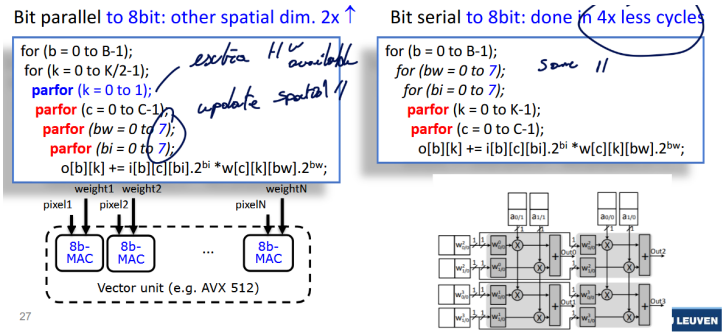


Figure 17: Potential gains of sharing

5.3 SoTA example of Quantization

5.3.1 Envision (KU Leuven)

It is a mult-gating approach with precision scalability. There are 256 MACS in a 16 by 16 array.

```
for (c = 0 to C-1);
  parfor (k = 0 to 15);
    parfor (b = 0 to 15);
      o[b][k] = i[b][c]*w[c][k]

// Can also be acting 512 MAC units and so on...
for (c = 0 to C-1);
  parfor (k = 0 to 15);
    parfor (b = 0 to 31);
      parfor (bw = 0 to 7);
        parfor (bi = 0 to 7);
```

This was successfully taped out and tested. They managed to adapt the supply voltage to gain even more in power efficiency.

5.3.2 Tensor Cores (Nvidia)

Same ideas occurs here where we can sort of extend the tensor in multiple dimensions if operating different datatype.

5.3.3 Hybrid training / inference chip in 7nm (IBM)

IBM has been pushing for lower and lower order of quantization. Proposing special structure for LLM inferences. They have some multi-precision compute array MPE with 16-way hybrid FP8 and 8-way FP16 engine in it. They do the same for int4 and int2 but they separate fp and int as they witnessed a significant energy saving and slight area reduction.

5.3.4 Binareye - digital and MS (KU Leuven & Stanford)

This extreme 1-bit quantization where we can use XOR gate to compute the multiplication. This is highly efficient in area and energy. The main drawback is the gathering of all those results which constitute the main bottleneck of this architecture.

The results were quite impressive for such low-order operations. This opens the possibility to run lightweight model on very efficient hardware.

We have really good weight stationary and we keep feeding input in parallel. We break down the input into Fx*Fy and repeat the feeding process for x*y cycles.

```
for (k2 = 0 to K/64-1); //for each output channel
  for (x = 0 to X-1); //for each in/output column
    for (y = 0 to Y-1); //for each in/output row
      parfor (c = 0 to 255); //for each input channel
      parfor (fx = 0 to 2) ; //for each kernel row
      parfor (fy = 0 to 2) ; //for each kernel column
      parfor (k1 = 0 to 63); //for each output channel
      o[b][k1+64.k2][x][y]+= i[b][c][x+fx][y+fy]
      * w[k1+64.k2][c][fx][fy];
```

5.3.4.1 Reducing the gathering bottleneck We can replace those large neurons array by a Mixed Signal approach that is composed of a switched-cap adder (sorts of ADC).

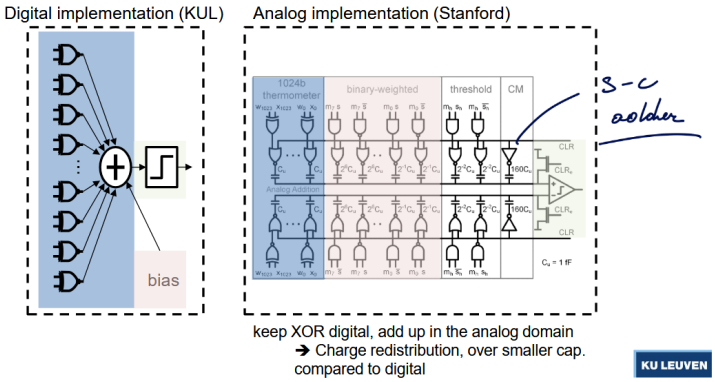


Figure 18: The switched cap implementation

This presented multiple issues:

- Variations of cap: using cap is better than a resistor ladder, … but high variation could make accuracy plummet
- Noise: again sufficient margin must be taken and extra technique should be employed to reduce impact of the noise
- Comparator offset: this can be quite detrimental but at least we can calibrate this effect reducing its impact

Even with this, the results are still stochastic around the perfect digital model. The energy used by the neuron array is reduced by a factor 12.

5.4 In memory compute

5.4.1 Concept

Instead of transferring all the time the data from memory, bring the computation in memory. The idea is to use the bitcell and use the selection line as the input and the weight are saved in the bitcell (weight stationary). So the data lines are actually the output lines.

By measuring the amount of discharge, we could be able to compute the result and digitize it.

5.4.2 Analog IMC

This seems like a good idea on paper but the amount of extra hardware around is important. We need to use DAC to convert the inputs in analog domain, sum up the current and digitize it with ADC. Finally some post process can be applied (ReLu, …)

Typically spatially unroll C*FX*FY in vertical direction. Unroll K in horizontal direction, B,X,Y temporal

5.4.2.1 Input pulse width Use bit width pulses to discharge a certain amount each time.

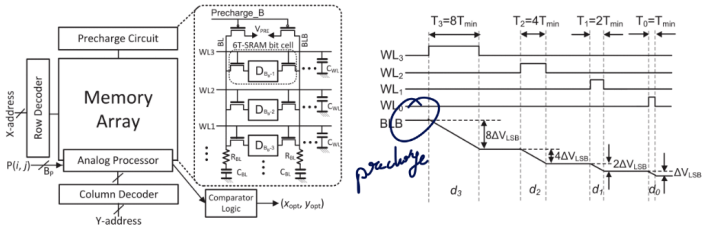


Figure 19: bit width pulses

We have superposition of binary weighted bitline discharges across multiple rows. Essentially a MAC operation with digital inputs and analog output. The inputs are now digital.

Quite nonlinear & random, difficult to manage large mismatches between SRAM cells

5.4.2.2 Input pulse count Here, all the pulses are the same, the only difference is the amount of pulse. Current-based addition still present large bitline non-linearity. This can be adapted using some non-linear modeling to counter-act it.

Both techniques use memory element as *current-source like* and the summation is realized with precharge-discharge cycle.

5.4.2.3 Voltage We use a regular resistance with ReRAM (or Flash) cell. We make resistive weighted sum.

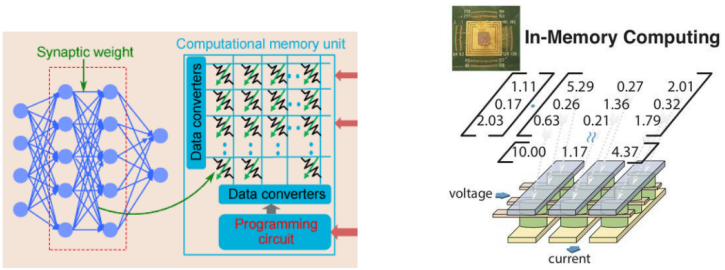


Figure 20: Voltage based IMC

This is kinda like the PE unit of google where we could save full the weight and directly route on the chip the data to go through multiple layers.

- THE GOODS:
  - Analog in-memory compute allows extreme input and output reuse
    - Due to compact memory (multiplier cell & added) & large arrays
  - Analog in-memory compute enables extreme weight stationarity
    - Program weights once, and leave them there forever?
    - If weight reloading is rare (needs large arrays, or small networks…)
      - Throughput/area increase (dense computing)
      - The larger is the memory the higher is the benefit. But not every NN will be this large to actually witness any gains.
- THE BADS:
  1. Can all NN layers exploit large arrays? → underutilization!
  2. Only precise at low quantization level? → Chip-specific training? Accuracy loss?

5.4.3 Digital IMC

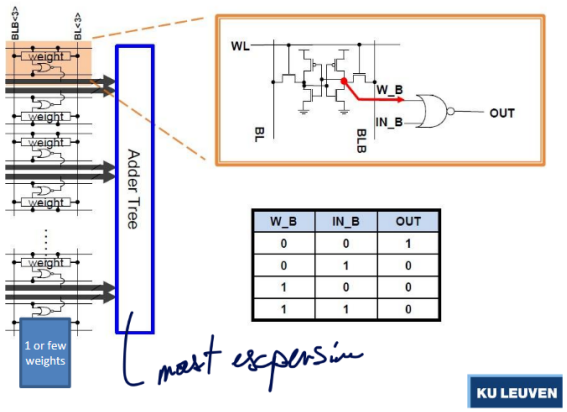


Figure 21: Digital IMC architecture

We digitize the 1*1 bit results and accumulation is done digitally. This adder tree is now the bottleneck. 1 weight per adder or weight bank. Here, results are deterministic which is better suited for the industry.

For example, TSMC provides special node for this kind of IMC. They share multiplier across 4 weights, it's not really IMC in the strict sense.

More freedom:

- Type of memory cell (now SRAM)
- Number of bitcells (weights) per multiplier $\rightarrow$ local data reuse (higher level stationarity)
- Number of cells per core (=size adder tree), or reconfigurable?
- Number of cores
- Data sharing between cores

Benefits of analog in memory compute still hold (but less!)

- In-memory compute allows extreme input and output parallelism (reuse)
- In-memory compute enables extreme weight stationarity

Compared to analog:

- More flexible (reconfiguration of data reuse in function of layer topology?)
- More reliable (precision guaranteed)
- But… less dense (Tops/mm2 lower)
- But… less efficient (Tops/Watt lower (at core level, not at system level?))

6 Lecture 6: Efficient deep inference: Sparsity, Scheduling, Fusion

6.1 Exploit sparsity

6.1.1 What is sparsity? Types of sparsity?

Not all NN are fully-connected NN. Thus, not all inputs go to every single nodes which result in matrices with empty spot. This is what we call **sparsity**. But not only the architecture affect sparsity but also: quantization (small values get rounded to 0) and explicit training rules (think about $ReLU$ function).

From this 3 types of sparsity emerges:

Type	Description	Advantages
Unstructured	Weights are distributed at random	Easy for software ppl, hard for hardware
Structured	Weights are pack in sparsity block	Easy for hardware ppl, hard for software
Density block bound	A ratio of weight/elements can be garante per row/column	middle ground

6.1.1.1 At training It is possible to only retain the dominant weights (can have some applications in specific AI cases). We add an optimization constraint which will also improve training efficiency:

$$\min_{\beta \in \mathbb{R}^p} ||y - X\beta||_2^2 + g||\beta||_1$$

Unstructured sparsity showed often better result but it is possible to obtain acceptable results with DBB sparsity. Sadly, the more recent models are less sparse since they are more complex and features are not easily detectible as it used to be in Resnet and other models.

6.1.2 Exploiting sparsity in memory size/access

The goal here is to avoid useless 0 data as it will hinder the bandwidth. Many encoding schemes exist depending on the sparsity (Compressed sparse row, bitmap, ...)

Using structured sparsity, it is much easier to setup a mask/index matrix and to efficiently store and deploy. With DBB sparsity, we need as many numbers as non-zero data BUT the matrix is regular contrary to unstructured data which can be challenging in modern libraries such as Numpy which only accepts regular matrices.

6.1.2.1 Envision processor They use some Huffman encoding inside the DRAM and DMA (huffman tree), it will analyze and find the useless operation to reduce data transfer.

6.1.3 Exploiting sparsity in processing

As seen previously, GPU relies on the fact we can stream lots of data in parallel and do the same operation. But having non-rigorous data may compromised the benefit of GPU.

Sparse BLAS libraries exist but need huge sparsity to have any benefits.

6.1.3.1 Envision processor On top of the memory flag to guard read as introduced previously, we re-use this flag to guard MAC execution. It only improves the power consumption but doesn't improve speed or throughput!

6.1.3.2 Sparse CNN engine There is some specific hardware tailored for such operation. It stores inputs and weights compressed and schedule **dynamically** the operations. This require an extensive control on operation and scheduler.

Does perform better than Envision at low sparsity $< 60\%$ due to the massive overhead. But this proves, dedicated hardware yield significant benefits!

6.1.3.3 Systolic array Using DBB, we can have a dedicated X MAC array per block which will be sure to have some operation to do at all time. We have the same throughput for less MAC.

6.1.3.4 CUDA On CUDA, there is now (Ampere) the cuSPARSELt library that assumes 2:4 DBB. It will force in a mux 4 operands by using non-zero indices MUX. At a perfect 50% sparsity we can almost double the operations!

6.2 Operator fusion / scheduling

It is important to understand the advantages and limitations of algorithms and hardware to better design them.

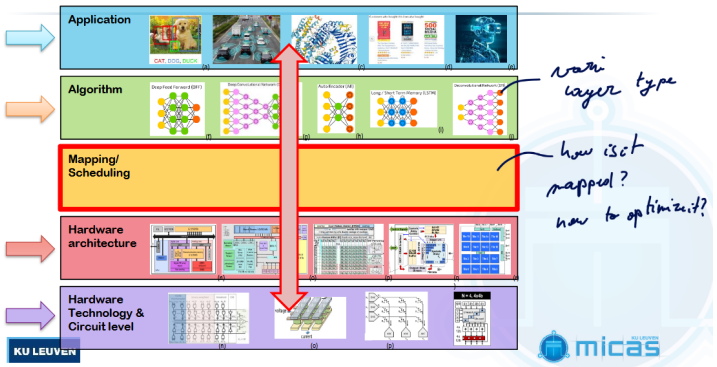


Figure 22: Better mapping leads to better performance

6.2.1 Single core / single layer

Besides the usual `for`-`parfor` as seen previously, we can leverage from various level of cache to hide the latency of accessing memory.

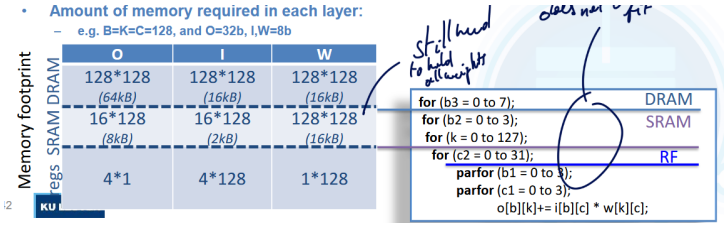


Figure 23: Memory level

A good and simple algorithm for scheduling and putting the right boundaries is to:

- If we have not enough memory a level x
  - Drop upper memory boundary
  - Cache size related
- Too many memory accesses of level x
  - Increase lower memory boundary
  - AI mem accesses/clock cycle

6.2.1.1 Automated performance estimation We can also think about possible automated tools that can find the best architecture for a given algorithm. The best is to build an *analytical performance model* that do not need extensive compilations or simulations for each NN performance evaluation.

This is what ZigZag is solving where based on:

- NN workload
- Mapping

- Hardware architecture & constraints
- Technology characteristics

It builds a cost model.

6.2.2 Single core / multi-layer

If we did our job well, we should have optimized single layer. But if we don't look at the interconnect between layers, we may miss potential benefits.

Typically in LLM, there is many linear layer, we could fuse the instructions together to reduce redundant architecture.

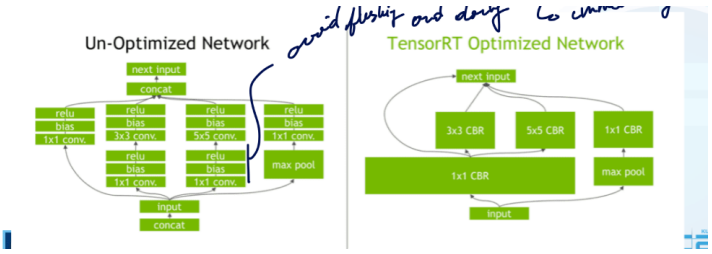


Figure 24: Tensor RT Optimized

As in threading, we can try to do some round-robin scheduling and have a divide and conquer approach. Instead of waiting for full output, we can start the next layer with partial output. But this is not possible for every architectures.

Split operation is really interesting for memory constrained architecture where we must rely on tiling or other operations to obtain the outputs.

6.2.3 Multi-core / multi-layer

6.2.3.1 Homogeneous We simply duplicate the core multiple time. It is easily scalable and it gives some mapping flexibility. But only allows output re-use.

6.2.3.2 Heterogeneous - Diana chip We have a distributed memory hierarchy with digital and analog cores. We can do pipelining and fusion with a small memory which allows for good performance and low shuffling of data.

It is flexible, maybe too flexible and requires tool to efficiently schedule (stream).

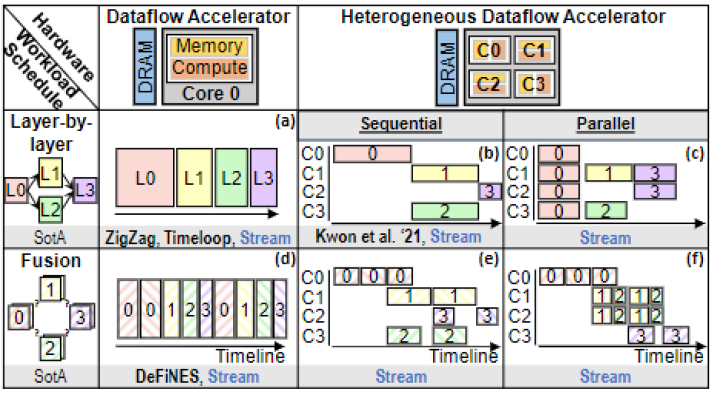


Figure 25: Scheduling across multi-cores

7 Lecture 7: Heterogeneous processor co-operation

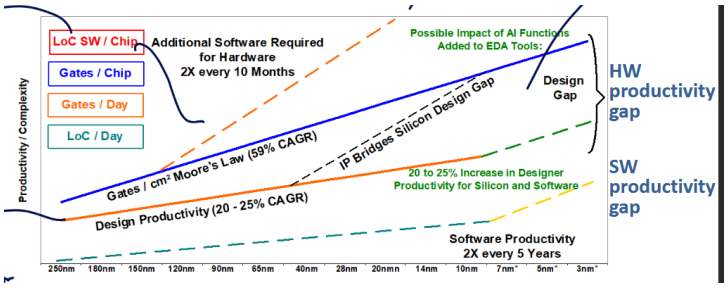


Figure 26: We need more productivity to keep on designing more complex chips

We will be looking at how we can keep up in productivity.

7.1 Improving design efficiency

7.1.1 IP reuse

One of the first way to improve productivity is to reuse IP. This technique has been extensively used for the past few years topping at around 90% of IP reuse in a 2018 chip. There is also an Open-Source movement for hardware like RISC-V.

The goal of RISC-V is to standardized and build a common API for Software and Hardware engineers to work on. This will allow for compatibility, shared compiler and simulators, …

RISC only provides the ISA **not** the full processor. It provides the format and set but also leaves room for variants and extensions. So, many flavours of processor exist each being tailored for a specific utilization.

7.1.2 IP extension

It is important to make them easy to adapt. Meaning, we should avoid writing low-level Verilog code but instead use some Hardware Construction Language like Chisel (Scala). The best choice is to use some OOP (object oriented programming) to be able to deploy and parametrized new modules easily.

With one object, we can produce many variants and insert it like we desire. From a dual issue to a quad issue, …

7.1.2.1 Changing the ISA On top of that, we can also support extra instructions not defined in the standard ISA. If we want to keep support with basic C-compiler we can only use the opcode `00010xx` or `01010xx`. For example, RI5CY allows for efficient hardware loop by introducing extra instruction.

AI-centered extension can be designed to support quantized values for efficient operations like in XpulpNN.

In XpulpNN, we provide some extra unit with multiplier designed to work at lower bit precision. We must start playing with VDD due to the modified critical path. The energy per operation is getting much better.

7.1.3 IP integration

We can also do some “Frankenstein” cores by mixing and integrating them together. Using PULP, we can easily mix and match components.

7.2 Improving deployment efficiency

The investement made for each new process node has a major software part. The hardware could help at hiding the complexity for software (choosing the right cores, …)

7.2.1 CPU-centric

By using multiple level of cores, we can tailor each of them for specific tasks and performance, maximizing their efficiency. This is like the Big.LITTLE architecture from ARM. Each core has the same ISA and can easily transfer workload from one core to the other.

We use a cache coherent interconnect so every cores have access to the same content at all time. Need some good cache coherency protocol such as bus snooping, MESI protocols, …

To monitor the workload, we monitor the *running average* of the task state (is the core used or not), if it crosses a certain threshold it could be migrated (**up migration**). The **down migration** threshold is lower than the up migration to avoid constant workload migration.

Table 8: Migration architecture

Architecture	Description	Advantages
Core pairing	We have a pair of big.LITTLE cores. Using DVFS (of core) we determine which core is active.	Quite simple to program and migrate.
Global task scheduling	Every core can migrate to any other core. We again use DFVS (of threads) to monitor usage.	Harder to monitor and dispatch the work.

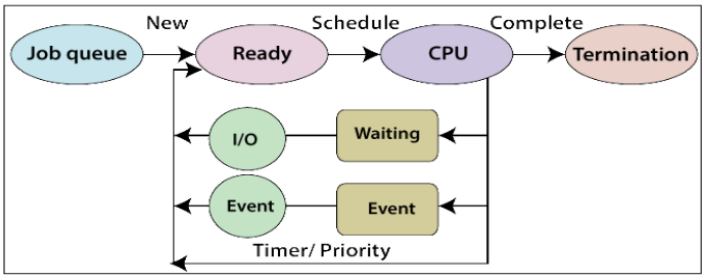


Figure 27: Typical scheduling

7.2.1.1 Scheduling Will try to maximize usage by looking at the state of the threads and checking if they are terminated. Scheduler is an art in itself and various level of hierarchy exist in schedulers.

But one key components is the need to synchronize them together to share data or commit data.

- data synchronization:
  - Avoid double access/parallel access. One thread locks the ressource, commit on it and release it. This ensures a chronological succession.
- event synchronization:

  - To make sure multiple thread end at the same time. Typically if we want to synchronize the output of a parallel function. We put a barrier to block on a loc.

Barriers can be realized in SW or HW! In SW, we have a lock that decrement the value and all thread must force a fresh read (no compiler optimization `volatile`) until the value of 0 is reached. This adds a lot of overhead and busy waiting.

In HW this can be solve with dedicated register such as the **Control Status Register**. It requires some special instruction (to make it atomic) but any core can access this CSR.

7.2.2 Heterogeneous SoC's

It is still a field that is not mature yet and many architecture and design variants are proposed to overcome some shortcoming.

7.2.2.1 hUMA - heterogeneous Uniform Memory Access We have only one shared memory (system) for CPU, GPU, … We can also make it virtually shared so the OS does the heavy lifting. Avoid snooping since it will have a lot of traffic. Typically the M1 of Apple.

The DRAM is IN the package and unified which allows for drastic performance boost.

7.2.2.2 Shared ISA for CPU, GPU, TPU, . . . Not solved yet but many obstacles. We don't want to go back to a non-RISC set where we have too many instructions. But, we are looking at an *unified intermediate* representation. So could be easily mapable to any device and the hardware only needs to compile it once and dispatch on the fly.

This is still quite abstract. The backend for support on hardware still requires lots of improvements.

Every compilation target still needs own (custom written/optimized) "finalizer" (optimized kernel functions). Which is not easy to add new/own HW accelerators.

- Difficult (impossible) for tools to automatically allocate code to cores
  - Detect / optimize fused kernel opportunities, …
- Difficult (impossible) for tools to automatically optimize schedulingacross cores
  - Detect / optimize parallelization opportunities, …

Current vendors just have hardcoded kernel implementations, lots of work to do until reaching this shared ISA.

8 Lecture 8: Adaptive SoC's

We must adapt at all time as the workload on a computer changes rapidly. The core workload changes, the frequency changes at all time, …

8.1 Open loop solutions for handling workload variations

8.1.1 DVFS

As seen in previous classes, dynamic voltage and frequency scaling is one of the most important tool for efficient and dynamic computing. By scaling the Vdd at runtime, we can make sure our processor is much more energy efficient thus less heat dissipation. $P_{dyn} = \alpha \cdot C \cdot f \cdot V_{dd}^2$

We can have preset pair of (f, V_{dd}) called **power states** and this can be chosen by the scheduler. One easy way is to use some LUT.

8.1.2 Problems of DVFS

It does not take into account slow runtime variations and it is too slow for fast variations. It is a feedback system which can have quite significant delay compared to small burst.

8.1.2.1 Slow variation It is a fixed LUT realized at production time, thus overtime those values can drift significantly. It doesn't control well the temperature either. That's why we need a closed loop system maximizing performance under thermal limit.

8.1.2.2 Fast variation When using intensively the core, we can witness *voltage droop* that can further impact performances beyond a certain threshold.

Typically, those droop can create setup time violation as the data may arrive late. The issue is that simply adding guardband can be problematic. Especially at lower power mode, since we are operating near V_{th} thus the guardband must be larger. We could also add some decoupling cap but this would cost lots of area and energy.

8.2 Closed loop solutions for handling variations

8.2.1 Resiliency $\rightarrow$ detect error & instruction replay: tolerate errors

We want to detect and correct errors due to dynamic variations.

8.2.2 Adaptivity for slow changes $\rightarrow$ AFS & instruction throttling: avoid errors

8.2.2.1 Error-Detection Sequential - EDS - Insitu This is the idea of the razor latch has seen in other classes:

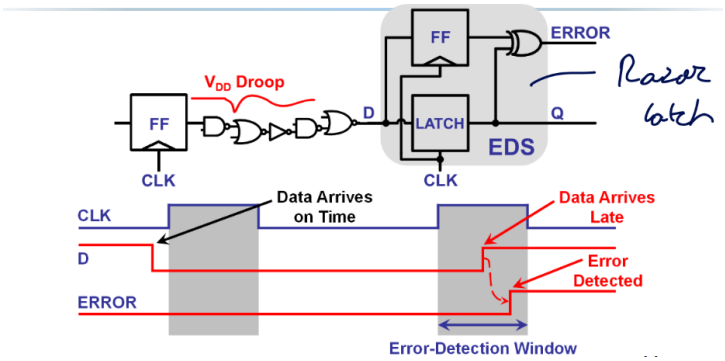


Figure 28: EDS

The error-detection must be carefully chosen. Too small and we could miss errors, too big and we could interpret some combinational logic change as a late data arrival while it is just data already ready for next clock cycle. To help with the last type of error, we must add min-delay penalty to avoid those errors.

8.2.2.2 Tunable Replica Circuit - TRC - Non Insitu We have a succession of gates, NOT, … that has an equivalent delay as the path we are actively monitoring. This TRC must be calibrated to make sure we are accurately monitoring the critical path.

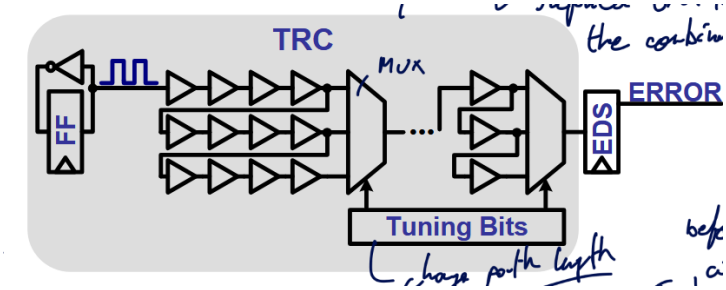


Figure 29: TRC

We also need some tuning bits to change the path length. We also need to make sure the TRC fails at any VDD, temperature, … before the critical path does.

8.2.2.3 Error recovery

Feature	Local: Value Injection	Local: Instruction Replay
Detection Method Compatibility	**EDS**	**EDS** or **TRC**
Error Feedback Mechanism	Value is injected back into the pipeline. If an error is detected, the value is fed back using a **MUX** (Multiplexer).	Uses a **flag** that propagates through the pipeline stages.

Feature	Local: Value Injection	Local: Instruction Replay
Pipeline Recovery Action	Requires stalling the **full chip for one cycle**.	At the WB stage, the pipeline stages are **flushed** and the instruction is **re-fetched** from where the miss occurred. Can use an Error Control Unit to replay at different (f,VDD).
Immediate Stall Required?	**Yes** (full chip for one cycle).	**No** immediate stall, but has a **significant delay** due to flush and re-fetch.
Implementation Status	**Never really implemented** in commercial products.	Requires tracking the **critical path** for each pipeline stage.

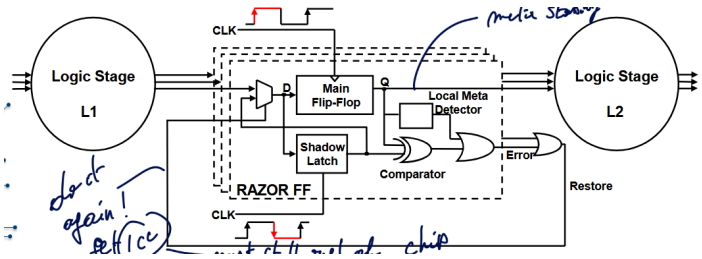


Figure 30: Local error recovery

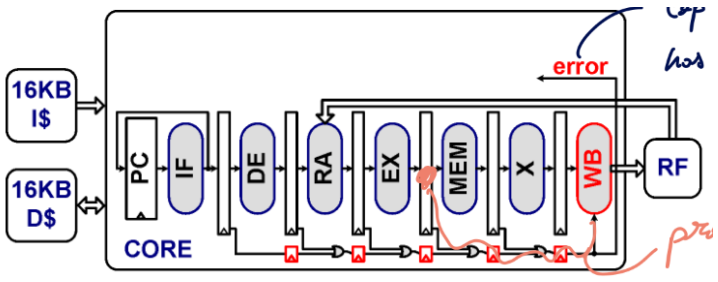


Figure 31: Instruction Replay

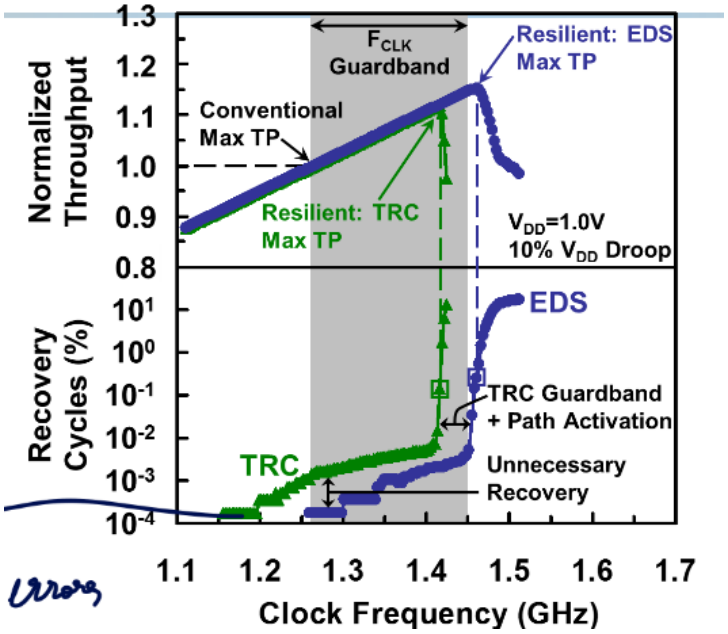


Figure 32: Gains: 16% TP w/ EDS - 12% TP w/ TRC

8.2.2.4 TRC Vs. EDS This difference is due to the fact that for TRC, extra guardband must be used.

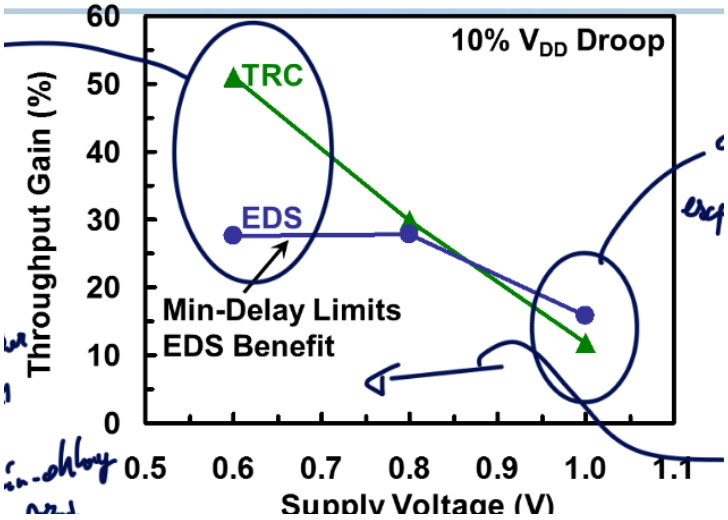


Figure 33: Throughput for various VDD

This graph, however, shows a totally different story! The issue lays in the fact that EDS has a minimum detection window for error detection. Remember, this window must be large enough and the path of fast path adapted for it. For TRC, since we are mimicking the critical path behavior, no need to have a constant guardband like in EDS.

Those solutions are easy to implement and quickly act upon error, but yet no system to actively retry without trying with the same settings.

Will ensure to change operation settings to not keep doing the same errors. We can use some thermal, aging or Vdd sensor to base our changes on them.

8.2.2.5 Instruciton throttling Fetch less instruction per CC and let the power decrease on its own. But after the throttle mode finish, they will again spike up! Sometimes, to avoid to let all cores run fast after a throttle, we can use **staged throttle** to gradually release throttle.

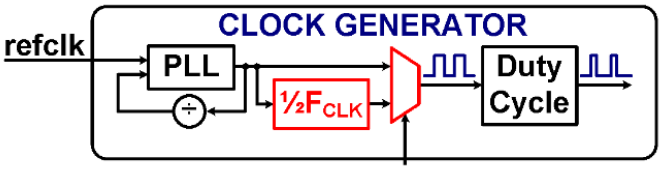


Figure 34: Adaptive frequency scaling

8.2.2.6 Adaptive frequency scaling Reduce the clock frequency with the same VDD as it is faster than changing VDD. We can imagine the mux being controlled by a Error Control Unit based on TRC.

This is easy to implement in commercial product. The main issue with those two methods is that fact it will still take some time to recover + propagation may be too long. We loose some clock cycle and so on. The droop is mitigated but not stopped.

8.2.3 Adaptivity for fast changes → adaptive clocking: avoid errors

The VDD droop will impact datapath and clock distribution timing. Which means the clock and datapath will slow down at the same time! Then, when it ramps up the delay goes faster and clock compress.

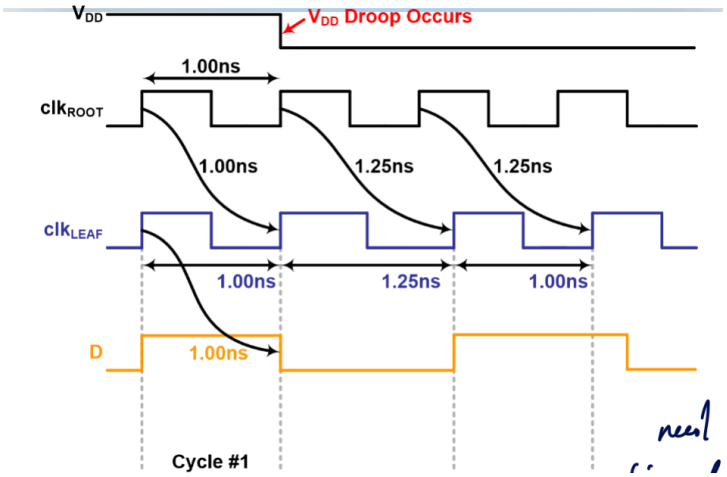


Figure 35: Compression and Expansion

At first, the data is compensated, but then as it compresses in the last clock cycle on the graph, the data fails and an error occurs. The stretch period must be long enough to allow other adaptations to kick in.

So why not extending this effect to allow more margin during the droop? This is **adaptive clocking distribution**.

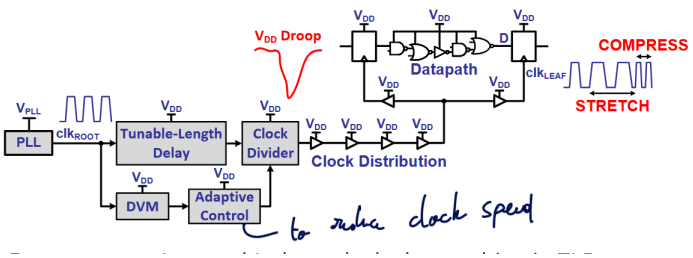


Figure 36: ACD

For this, we also need some Dynamic Variation Monitor (TRC) and use adaptive control, clock divider and a tunable length delay.

The top path is the fast response with clock stretching while the bottom is slower. The TRC, the slow path, triggers clock frequency adaptation in adaptive clock system.

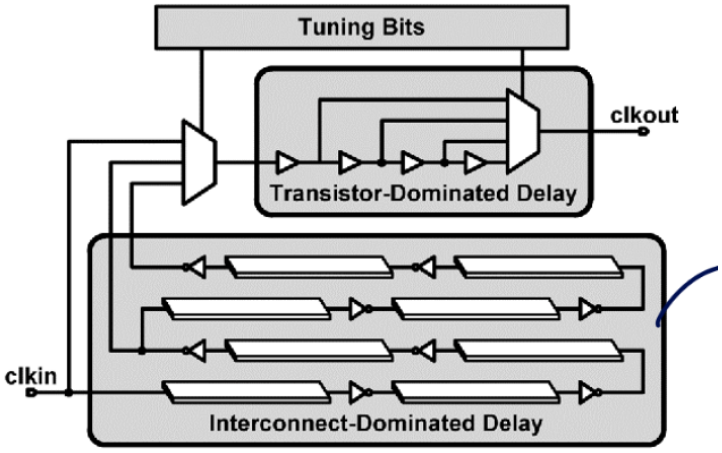


Figure 37: Tunable Length Delay

8.2.4 Future → UVFR

We want to add some feedback for the droop preventions to better monitor and adapt.

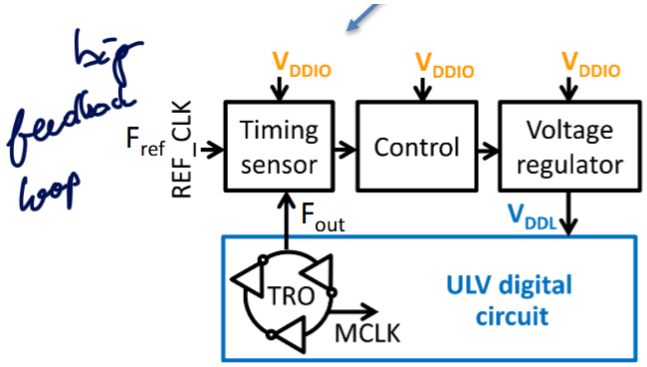


Figure 38: Unified Voltage & Frequency Regulation (UVFR)

We adapt the clock itself, linked with VDD droop in a single control loop. LDO regulator controlled on average (Fref - Fout). The clock is generated on the same supply voltage as logic so will follow together the speed.

UVFR avoids timing-margin violations but is not yet very programmable. But it adapts vdd to temperature, die-to-die and ageing variations.

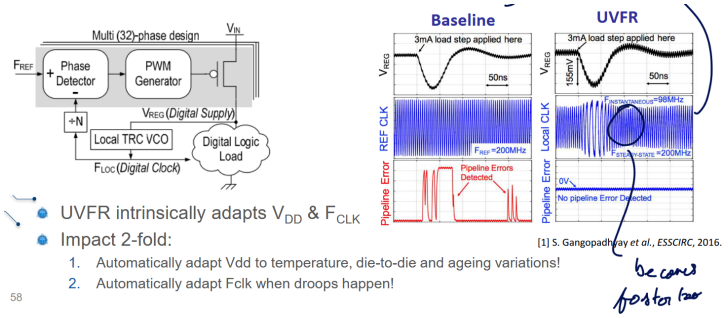


Figure 39: UVFR

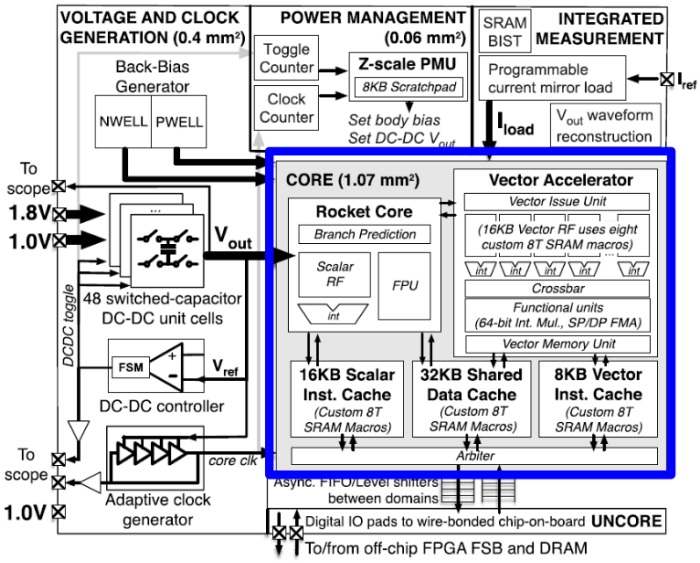


Figure 40: RISC-V example

8.2.4.1 RISC-V example The voltage regulator is implemented using switch-cap voltage regulator. By controling the toggling and which array to use, we can produce various voltages. The feedback circuit allows to precisely control the output voltage when it fluctuates. The power consumption governs the discharge rate and thus we must adapt the DC DC toggle rate to keep a sufficient supply. This creates large ripple on VDD and some IC's can be sensitive to it.

This is why we use an adaptive clock generator to avoid this ripple effect.

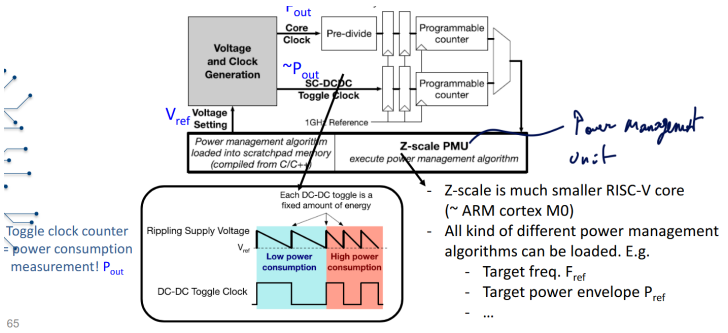


Figure 41: Power management

The idea is to no longer have fixed power states but set an average frequency.

9 Paper to Read

9.1 Lecture 1

> **Paper to read**
>
> *Paper1*: A New Golden Age for Computer Architecture J. Hennessy AND D. Patterson
>
> *Paper2*: Apple M1: Ditching x86 A. Frumusanu
>
> *Paper3*: Paper2: Apple M1 deep dive: Micro-architecture (pg2) Anandtech, Andrei Frumusanu; *link related to the article*

9.2 Lecture 2

> **Paper to read**
>
> *Paper 1*: Attention is all you need sec. 1-5

9.3 Lecture 3

> **Paper to read**
>
> *Paper 1*: How to keep pushing ML accelerator performance? Know your rooflines!

9.4 Lecture 5

> **Paper to read**
>
> *Paper 1*: ENVISION: A 0.26-to-10TOPS/W Subword-Parallel Dynamic-Voltage-Accuracy-Frequency-Scalable Convolutional Neural Network Processor in 28nm FDSOI

10 Questions

Access the online Google docs with this link.

10.1 Lecture 1

1. *What is Dennard's law and how is it linked to the evolution of computer architectures over the last 30 years? Describe the different phases we see in this evolution, and the architectural consequences. Illustrate this with examples from processors discussed in class and the papers to be read.*

> **To be answered**

2. *Why do Henessey and Patterson say it is a New Golden Age for computer architectures, and which opportunities should be exploited in this Golden Age, according to them? (See L1_Turing.pdf)*

> **To be answered**

3. *Discuss the different types of processors present in a modern embedded system (like iPhone's, smart glasses, Tesla cars, or…). In what sense do they differ? Why are they all there? How do they exploit area for efficiency? (after L8: How would they exchange data and processing jobs?)*

> **To be answered**

4. *Apply the different concepts from this class to the Apple M1 processor. Which "area for efficiency" techniques do they exploit and why? What is unique about the FireStorm cores. (See also paper L1_Apple.pdf)*

> **To be answered**

5. *What are the reasons for the going to multi-core CPU's, first homogeneous and later heterogeneous? How are the micro-architectures of the CPU's different/similar, and how are they exploited? Does this offer energy efficiency and/or carbon footprint benefits?*

> **To be answered**

10.2 Lecture 2

6. What are ISP's and IPU's? Explain the evolution in their architecture, using example processors to illustrate the trends. What are the main characteristics that distinguish the Google Pixel IPU from a CPU and from the Hexagon processor?

> **To be answered**

7. Explain the workload of embedded rendering tasks and how this leads to new GPU processing architectures beyond CPU's. What are the main characteristics that distinguish GPU's from CPU's? How do mobile GPU's differ from cloud GPU's?

> **To be answered**

8. What is the difference between a fully connected, a convolutional neural network layer and a depthwise separable layer. Which one is considered more hardware efficient, and why?

> **To be answered**

9. What is the attention mechanism in transformers, and what operations does it require in prefill and decode? what are its benefits and downsides compared to convolutional networks. (Also use L2_Attention.pdf)

> **To be answered**

10. (After having studied L3-5) How can certain types of neural network layers be more efficient from an algorithmic point of view and at the same time be more inefficient from a HW point of view?

> **To be answered**

10.3 Lecture 3

11. How can Rooflines be used to assess NPU performance? How do the techniques we discussed in L3-6 impact the rooflines themselves, or the position of a workload in the roofline. (also use paper L3_Rooflines).

> **To be answered**

12. [After L4:] What is the difference between spatial and temporal unrolling of for-loops, and how does it impact the arithmetic intensity? What sub-types of spatial and temporal unrolling can you distinguish?

> **To be answered**

13. Explain the difference between the GeMM execution dataflows of:
    1. Tesla's NPU
    2. An Nvidia TensorCore

> **To be answered**

14. How can spatial unrolling be extended from the datapath to the core level? What sharding opportunities exist and what are their benefits or downsides?

> **To be answered**

15. [After L4:] Given a nested (par)for loop representation (see examples in L4), explain me what the data parallelism and the stationarity is (spatial and temporal unrolling). Also derive the AI.

> **To be answered**

10.4 Lecture 4

16. What is tiling, how does it improve arithmetic intensity and how can you recognize it from a set of nested for loops?

> **To be answered**

17. [Also using input from L3]: For a given number of MACs (e.g. 256), how can respectively a 1D, 2D and 3D MAC arrays exploit spatial and temporal data reuse, and which of these do you expect to result in the most efficient datapaths?

> **To be answered**

18. What is a systolic array, and what are its benefits and downsides. Can the downsides be overcome? (Also use the content of L4_TPU.pdf)

> **To be answered**

19. What is utilization, and what aspects influence the utilization of an AI processor core?

> **To be answered**

20. Why does a (performance or energy) roofline actually encompass multiple roofs, and what does this have to do with temporal loop optimizations?

> **To be answered**

10.5 Lecture 5

21. What data types are relevant for ML computation? What are their main differences? How can this be exploited in an ML processor ans what is the impact on the roofline model? Also use paper L5_MoonsISSCC.

> **To be answered**

22. What different ways are there to build precision scalable MAC units / MAC arrays, and what is the impact on the nested for-loop representation? Use this to discuss the dataflow and utilization consequences (opportunities and challenges) of adapting the MAC precision?.

> **To be answered**

23. Discuss the parallelism, stationarity and sparsity aspects (after L6) of the Envision processor and the Nvidia Tensor Cores, and how these aspects depend on the chosen data type.

> **To be answered**

24. What are the opportunities that come from extreme binary quantization (in the digital, mixed-signal and the in-memory domain? You can use BinarEye as an illustration.

> **To be answered**

25. What is the difference between analog and digital in-memory compute and what are their relative benefits and weaknesses?

> **To be answered**

10.6 Lecture 6

26. What is neural network sparsity and how can it improve energy efficiency and/or throughput in neural network storage as well as processing? Include the relative advantages and disadvantages of more or less structured sparsity.

> **To be answered**

27. How can one analytically estimate the utilization (throughput) and energy consumption for executing a given neural network layer on a given hardware architecture?
    - Exercise: given a set of nested for-loops and a memory allocation (e.g. a variation on slide 41, with a different loop set), discuss required memory sizes, spatial utilization, memory bandwidth requirements, AI

> **To be answered**

28. How can one optimize a schedule and/or hardware architecture for a set of workloads.
    - Exercise: given a set of nested for-loops (e.g. a variation on slide 41, with a different loop set), discuss possible hardware, scheduling and/or memory allocation optimizations and their impact.

> **To be answered**

29. What is operator fusion (layer fusion) and why does it matter? What are its challenges?

> **To be answered**

30. What are the benefits and downsides of homogeneous/heterogeneous multi-core implementations and scheduling? Illustrate with the L6_Diana chip.

> **To be answered**

10.7 Lecture 7

31. Using slide 4, explain the causes and solutions to the HW- and SW-productivity gap.

> **To be answered**

32. What is RISC-V and how does it allow individuals to speed up SoC design, without giving up on customization? (discuss implementation flows and standard and custom extension approaches)

> **To be answered**

33. Which CPU extensions make sense in this era of embedded AI processing. Explain the impact of different opportunities on the system efficiency (also use L7_PulpNN).

> **To be answered**

34. How can hardware designers make heterogeneous multiprocessing systems based on CPU cores (big.LITTLE) easier to use/program? How is scheduling and synchronization organized in such systems? (see also L7_ARMscheduling)

> **To be answered**

35. What solutions are proposed to alleviate using truly heterogeneous multi-core CPU/GPU/NPU systems? What challenges remain?

> **To be answered**

10.8 Lecture 8

36. How do DVFS and AFS work, what is the difference between them and what are their advantages/disadvantages?

> **To be answered**

37. What is the difference between resilient and adaptive designs? Give and explain an example of both. Are they sometimes combined?

> **To be answered**

38. What is the difference between EDS and TRCs, and how can they be used in a pipelined processor? What are their advantages / disadvantages? Explain their different performance (L8, slide 29/30).

> **To be answered**

39. What can be done to overcome very fast voltage droops under a fixed supply voltage and static clock generation.

> **To be answered**

40. What can be done to overcome very fast voltage droops, when exploiting integrated supply and voltage regulation (illustrate with L8_Zimmer)

> **To be answered**

10.9 Guest Lecture

41. How does NXP's micro NPU differs from a more traditional NPU, such as e.g. the Envision processor. Why are they making these choices at NXP?

> **To be answered**

42. Explain the operation of the systolic dot product array of NXP for a GeMM workload. What are its benefits and downsides compared to a regular systolic array?

> **To be answered**

43. Which techniques are used in NXP's compiler use to optimize memory accesses or memory useage?

> **To be answered**

44. Apply the roofline model to the different phases of transformer execution. What is the impact of the datatype in this? What is the impact of the embedding length in this?

11 License

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